

Machine Learning-based Air Pollution Prediction Model

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Abstract— Air pollution is currently a critical issue for both public health and the environment. It is vital to provide advance notice of pollution levels, and air quality forecasts can play a crucial role in achieving this goal. To measure pollution levels, experts rely on the air quality index (AQI). For this research, gathered data on air pollution, specifically focusing on Fine particles (PM_{2.5}), Sulphur Dioxide (SO₂), Nitrogen Dioxide (NO₂), and Carbon monoxide (CO), which were used as the primary dataset. Different machine learning models, including linear regression, lasso regression, random forest regression, and K-nearest neighbor regression, were then employed to analyze the collected data. The mean absolute error(MAE), mean-squared error (MSE), root-mean squared error(RMSE) and accuracy are used to evaluate the performance of the ML model. The comparison between these models is also discussed in this paper. Random forest regression is giving the high accuracy and low RMSE among other models.

Keywords— Air Pollution Prediction, Feature Engineering, Air Quality Index, Machine learning Models

I. INTRODUCTION

According to the central environmental authority (CEA) in Sri Lanka, air pollution can cause several health effects such as headaches, vomiting, acid rain, dizziness, lung and heart diseases. In Sri Lanka, burning organic waste products from the agriculture industry and petroleum refining are the main sources of air pollution. According to data particulate matter with a diameter of 2.5 micrometers or less than 2.5 micrometers, is highly affected to human health [2]. There are different gasses that cause air pollution. So, we need one index for our research to train the ML model. Because of that, we have calculated Air Quality Index (AQI) from the all gasses which we collected. So here we consider these four major pollutants which are nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), and the leading pollutant fine particulate matter (PM_{2.5}). Each pollutant has a sub-index and scales at a different level. The AQI uses a scale from 0 to 500 to evaluate the quality of the air, and it includes six levels ranging from "Severely polluted" to "Heavily polluted," "Moderately polluted," "Lightly polluted," "Moderate," and "Good." [3]. The purpose of these levels is to indicate the potential impact on human health and offer a numeric point of reference for individuals engaging in outdoor activities. With the calculated sub-index of each pollutant, we finally take the maximum sub-index as the Air quality index.

Before the rise of machine learning, people were used to probabilistic functions and previously collected data statistics to predict air pollution. But with the advancement of machine learning, training highly accurate ML models which work on probability functions and provide good

predictions were implemented for everyday operations. The use of machine learning has demonstrated remarkable resilience in predicting natural phenomena. Therefore, the application of Machine Learning (ML) techniques for Air pollution level prediction is explored in this research.

The prediction of the AQI was carried out in this paper using supervised machine learning algorithms. Supervised learning includes a range of algorithms, such as Random Forest, Naive Bayes, Nearest Neighbor, Kernel SVM, SVM, and Linear Regression[4]. From there, we selected these regression algorithms which are Random Forest, K – Nearest Neighbor and linear regression and Lasso Regression algorithms. To evaluate their accuracy, the RMSE, MSE and MAE were calculated for each of these ML Models. Python, Pandas, Matplotlib, and Sci-kit Learn were utilized in the implementation of these algorithms. The primary objectives of this research are to improve the procedures that have been used to predict AQI, to enhance our knowledge of AQI, and to understand the effects of poor air quality. This research's result, the Air Quality Index and AQI bucket, allows individuals to know how polluted the air is before they approach a city. After the COVID-19 situation and an increase in the number of machinery and vehicles that release carbon dioxide into the air, mainly in the Colombo area, the AQI level has consistently exceeded unhealthy levels. Therefore, reducing air pollution in Sri Lanka is the key goal of this paper.

II. LITERATURE REVIEW

In this section II discusses the literature review under two main categories which were past related research on the air pollution systems and air pollution recording system in different countries.

A. Related work on Air Pollution Prediction using Machine Learning

The authors of [5] introduced an air quality monitoring system that included both an assessment module and a forecasting module based on input features such as pollution levels, weather conditions, and chemical components derived from the WRF-Chem model. Using various groups of features and classification algorithms such as linear regression, SVM and random forest, the authors conducted experiments on 74 cities in China. Their findings demonstrated that the combined model yielded more accurate predictions than the individual models.

In [6] the authors proposed an air pollution monitoring model for determining air pollutants using a data mining algorithm. The model was developed to measure the presence of CO, NO₂, SO₂, and O₃ gases using a smart sensor micro-converter that downloads the pollutant levels to

a network-connected personal computer. The system takes into account parameters such as MQ5, MQ7, temperature, and humidity datasets to calculate the pollutants in different areas. The model could also forecast future pollution levels using the ID3 algorithm, which calculates entropy and information gain. Based on this information, the system provides an alert message to warn users about potential pollution.

In [7] the authors reviewed research studies that employed big data analytics and machine learning models to evaluate air quality and identified potential challenges and resource constraints for future research. The article also highlighted the issue of hardware limitations affecting the accuracy of air quality monitoring and assessment. The authors proposed that in order to address this issue, further research should be conducted on data quality modeling and automated real-time validation.

In [8] the authors proposed a prediction module and air dispersion module using the Spyder integrated development environment in Python. They tested the efficiency of several machine learning algorithms, such as K-Nearest Neighbor, Support Vector Regression, Random Forest, Multi-linear Regression, and Multilayer Perceptron Regression, on a given dataset. The authors utilized root mean squared error (RMSE) to assess the results of these algorithms. Based on their findings, the Multilayer Perceptron algorithm yielded the best outcome, displaying the lowest RMSE in comparison to the other algorithms.

The authors of [9] proposed a machine learning-based air pollution monitoring system with a focus on $PM_{2.5}$ prediction accuracy. The objective of their study was to assess the effectiveness of different machine learning models in forecasting elevated levels of pollution. Linear regression, ANN, and LSTM models were employed and the findings indicated that the LSTM model yielded higher accuracy in predicting elevated $PM_{2.5}$ values compared to the other models, with reasonable precision. However, the authors also noted some issues with the LSTM model.

B. Air Pollution Recording Systems

The Central Environmental Authority of Sri Lanka has been actively monitoring the air quality in the country since 2013. In order to facilitate air quality management, raise awareness of the public, address health concerns, conduct trend and status analyses, and share relevant data with decision-makers, two automated ambient air quality monitoring stations have been set up in Battaramulla and Kandy. These stations were established in accordance with the national siting criteria.

These monitoring stations measure six different air pollutants, including fine particles ($PM_{2.5}$), sulfur dioxide (SO_2), coarse particles (PM_{10}), nitrogen dioxide (NO_2), carbon monoxides (CO), and tropospheric ozone (O_3), on a regular basis. The results are used to produce an AQI that is made available to the public every day, providing information about air quality and related health concerns in the affected areas. In addition to measuring air quality, the monitoring stations also measure the speed of the wind and its direction, as well as relative humidity (RH) and temperature.[10].

In India, [11] ambient air quality is monitored by various entities including the State Pollution Control Boards

(SPCBs), Central Pollution Control Board (CPCB), Pollution Control Committees (PCCs), and the National Environmental Engineering Research Institute (NEERI). In all, 523 manual monitoring stations are already in use across the country. They also use a variety of measurement procedures and instruments to provide varying ambient concentration results. Many systems are primarily used to monitor the pollutants $PM_{2.5}$, SO_2 , and NO_2 during a period of 24 hours.

In China, the GAIA air quality monitoring stations employ advanced laser particle sensors to detect $PM_{2.5}$ pollution, which is considered one of the most detrimental air pollutants, in real-time. The system is read the daily AQI value based on the pollutants. It's also quite simple to set up, requiring simply a WIFI access point and a USB power source. Once connected, air pollution levels are displayed on their maps instantaneously and in real time [12].

Sydney [13] is Australia's largest city, is closely monitoring its air quality with over 50 monitoring stations managed by the New South Wales Office of Environment and Heritage (OEH). The OEH is responsible for ensuring the health of the environment and measures six air pollutants at these monitoring sites, including CO, NO_2 , O_3 , SO_2 , and two particulate fractions (PM_{10} and $PM_{2.5}$). As air pollution levels continue to rise, the OEH has implemented AQM 65, which measures PM_{10} , $PM_{2.5}$, O_2 , NO_2 , CO, and SO_2 . A study was conducted to test the accuracy and reliability of the AQM 65, and it was found that it could be used in conjunction with the current air monitoring stations.

III. METHODOLOGY

In this section III, discussed about the dataset we utilized, air quality index and system implementation.

A. Data Collection and Problem Statement

In this research, we have used a publicly available dataset from in Sri Lanka. The dataset has air quality dataset from January 2018 to December 2021. The dataset used in the study contains 29,532 instances and includes hourly averaged The monitoring stations in the Colombo area measure both the Air Quality Index (AQI) and the concentrations of $PM_{2.5}$, NO_2 , CO, and SO_2 . The data was obtained from the Central Environment Authority and the National Building Research Organization and was used to train the algorithm to identify air quality [14].

Here prediction output is air quality index. AQI value is a numerical value not a categorical label so in this case used regression algorithms to train the dataset.

The features of the data set were to be as follows:

- Date(D/M/Y)
- $PM_{2.5}$ Concentration(ug/m^3)
- NO_2 Concentration(ppm)
- SO_2 Concentration(ppm)
- CO Concentration(ppm)
- Air Quality Index (AQI)

B. Data Analysis

- *Pre-processing the data*

Measurements are sometimes unavailable due to a lack of measuring equipment, most commonly due to defective sensors or transmission failures, or a lack of required data points. So, we can't train an algorithm with the dataset's missing values. As a result, feature engineering or data pre-processing were employed to improve performance.

However, to improve efficiency, implemented the main three feature engineering techniques which are imputation, scaling and handling outliers. Here used boxplot visualization with pandas and seaborn to find outliers in the data set. From Figure 1 to Figure 5 shows the outliers of system features separately.

- *Boxplots with seaborn*

Compared to other statistical graph methods, the box plot is particularly effective in visually presenting outliers. Outliers refer to data points that lie far from the minimum and maximum values within a dataset. By highlighting outliers or unusual data points, the box plot allows us to quickly spot them and examine them more closely.

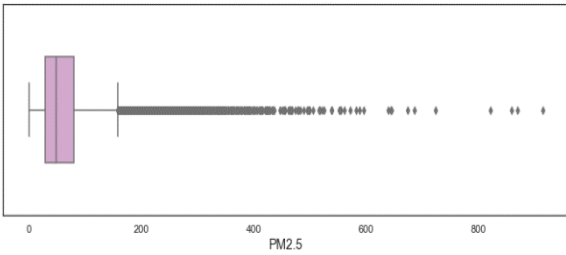


Fig. 1. Outliers of PM_{2.5} dataset

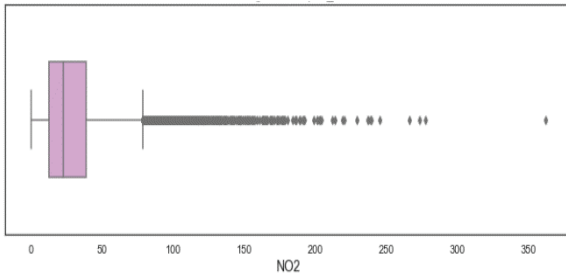


Fig. 2. Outliers of NO₂ dataset

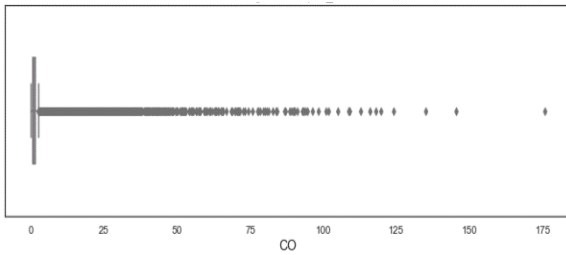


Fig. 3. Outliers of CO dataset

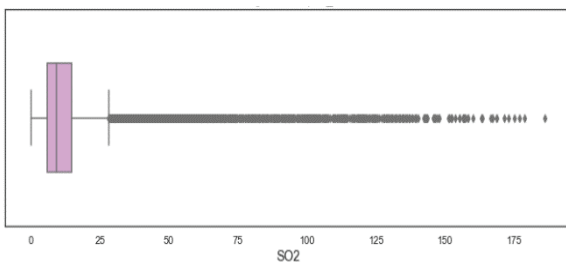


Fig. 4. Outliers of SO₂ dataset

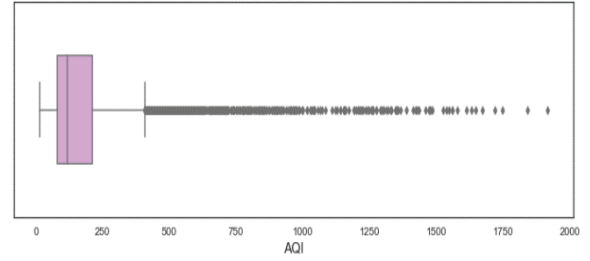


Fig. 5. Outliers of AQI dataset

C. Model Building

- *Air Quality Index*

The concentration of various pollutants is used to evaluate the level of air quality through the Air Quality Index (AQI). It reflects the level of air cleanliness or pollution and provides health recommendations depending on the AQI level. To determine the AQI, it is necessary to have data on at least two pollutants, one of which must be PM_{2.5}. If there is an absence of PM_{2.5} data, then the data is inadequate for computing the AQI.

We assume the AQI value as the maximum sub-pollutant index after calculating each sub-index. The sub-pollutant index for individual pollutants is calculated using from measured gasses in 24 hourly average concentration value of PM_{2.5}, NO₂, SO₂, and CO also 8-hourly in case of at the monitoring location [8] [11].

- *Training Model*

Once the data preprocessing and AQI value calculation are completed, the dataset is split into two groups: training and testing [5]. The proposed model's process is described in Figure 6.

Training phase: After analyzing the data in the dataset, the ML system employs the chosen ML algorithm to construct a model represented as a line or curve. In our case, we utilized 80% of the dataset specifically for training purposes.

Testing phase: The inputs are sent to the system, which is then checked for performance. The accuracy is evaluated. In testing, we used 20% dataset in all datasets.

These are the algorithms that are used for training the dataset.

- Linear Regression
- Lasso Regression
- K Nearest Neighbors Regression
- Random Forest Regression

The dataset was trained using regression machine learning algorithms, with default parameters used in each case. The implementation used Python-based libraries including sci-kit-learn and Pandas, and the PyCharm integrated development environment was utilized.

- *Model Evaluation*

Once the model training phase has been completed, the model is utilized to predict the AQI value using a preprocessed dataset. The best machine learning algorithm is chosen based on accuracy.

There are three widely used evaluation metrics for regression problems:

- The MAE (Mean Absolute Error) is determined by taking the average of the absolute values of the errors.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \tilde{y}_i| \quad (1)$$

- Mean Squared Error (MSE) is the average of all squared errors: it "punishes" greater errors and is frequently helpful in real world applications.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2 \quad (2)$$

- The RMSE, assesses the average deviation of predicted values from actual values by computing the square root of the mean of the squared errors, and it may be expressed in "y" units as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2} \quad (3)$$

Equations 1, 2, and 3 represent the mathematical expressions for the parameters, where y_i denotes the observed value of the i^{th} sample, $\tilde{y}_i$ represents the predicted value of the i^{th} sample, and n is the total number of observations considered. There are many research publications on machine learning prediction using ML algorithms, but many of them compare SVM, ID3, Random Forest, and regression algorithms using single pollutant based on the MSE and r^2 parameters. The author of [14] used two major pollutants SO_2 and NO_2 , and using a linear regression model to predict the results. The author of [1] used five machine learning methods and compared their results with the only parameter RMSE values. However, in this study, were used four major pollutants for prediction and four different machine learning algorithms. Specially three main feature engineering techniques were used for data pre-processing. The results were predicted based on the pollutants and compared and described using three parameters which are main parameters accuracy, MAE, MSE and RMSE to produce the best result for each of ML algorithms.

D. Machine Learning Models

• Linear Regression

Regression is a statistical approach used to model a target value based on independent predictors, often used for forecasting and determining cause-and-effect relationships between variables. The method of regression varies depending on the number and type of independent variables and the relationship between them and the dependent variable. The generic equation for linear regression is given as:

$$Y = A + B * X \quad (1)$$

In the above equation, the predicted value is obtained using A, B, and X. A is the intercept, and B is the coefficient derived by the regression model for the predictor present in the data. The performance of the Linear Regression model can be evaluated through the graph depicted in Figure 7, which shows an accuracy of 89.23%.

• Lasso Regression

Lasso regression is a form of linear regression that utilizes the "shrinkage" approach, where the coefficient of determination is diminished to zero. This technique aids in eliminating insignificant independent variables from the model, resulting in more precise predictions with fewer variables. The Lasso regression model's performance is also noteworthy, with an accuracy of 88.95%, as displayed in Figure 8.

• K Nearest Neighbor (Regressor)

The K-Nearest Neighbors (KNN) is one of the most popular and flexible machine learning algorithms, which is simple to understand and implement. The value of K specifies the number of neighbors considered for making the prediction. When it comes to measuring similarity, various factors can be taken into account, such as distance. The K-Nearest Neighbors (KNN) method is a feature similarity-based approach that can be applied to both classification and regression problems. The KNN model has shown an accuracy of 92.65% when $k=7$, as demonstrated in Figure 9.

• Random Forest Regressor (RF)

Random forests (RFs) are a type of ensemble learning method that can be employed for numerous applications, including regression and classification. In the training phase, Random Forests (RFs) create multiple decision trees and generate a class that corresponds to either the mode of classes or the average prediction of individual trees for classification and regression tasks, respectively. [16],[17]. The Random Forest model demonstrated an accuracy of 88.95%, as represented in Figure 10, which is considerably higher than other regression models.

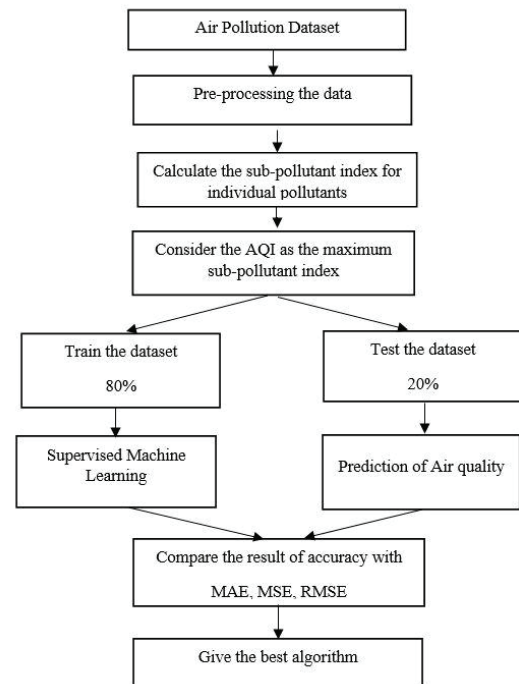


Fig. 6. Architecture of the proposed model

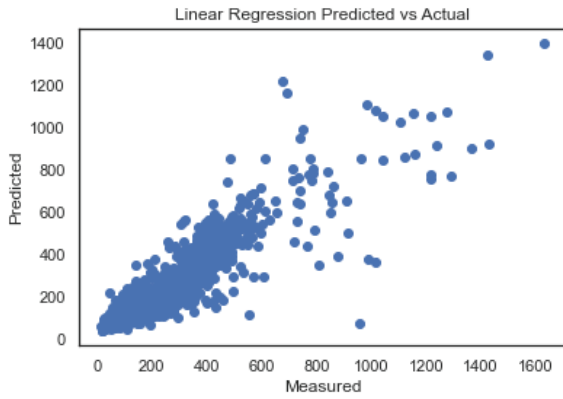


Fig. 7: Linear Regression Predicted vs Actual AQI Value

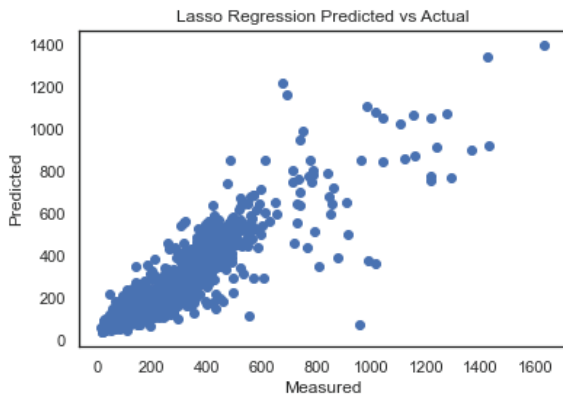


Fig. 8: Lasso Regression Predicted vs Actual AQI Value

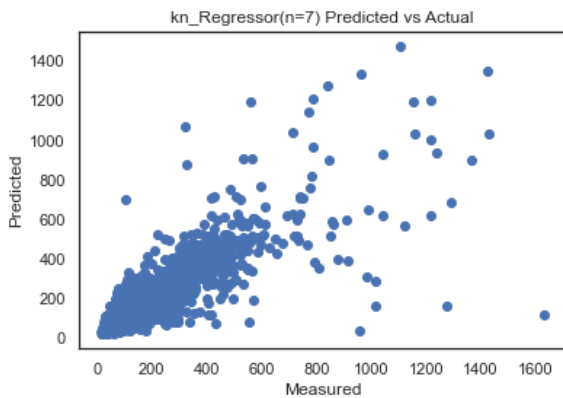


Fig. 9: KNN Regression Predicted vs Actual AQI Value

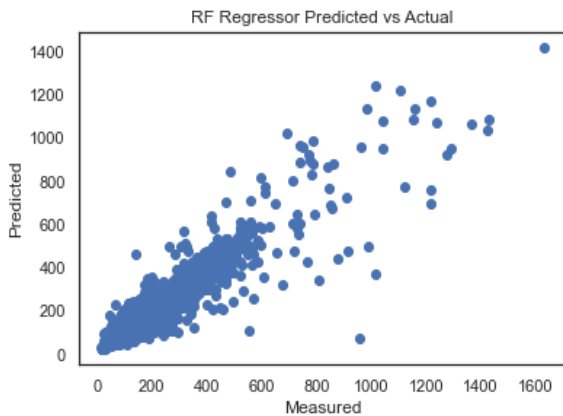


Fig. 10: RF Regressor Predicted vs Actual AQI Value

IV. RESULTS AND DISCUSSION

The research utilized four machine learning models, namely Linear Regression, Lasso Regression, K-Nearest Neighbor, and Random Forest Regressor to predict the AQI based on various features. Several metrics, including Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE) and accuracy were employed to evaluate the model's performance. In order to obtain a reliable regression model, a low RMSE value and high accuracy were considered crucial [15].

According to the Table 1, we have achieved 99.87% accuracy in Random Forest Regressor, since RF model gives the highest overall best accuracy and lowest RMSE value compared to the other models. The model that is best suited for this prediction process can be identified. In figure 11 shows the comparison of RMSE value of four machine learning models.

Many existing research that used KNN, Linear Regression, and Random Forest algorithms achieve improved accuracy. The machine learning algorithms' low accuracies may be attributed to the training dataset's incompleteness. The presence of missing values and noisy features in the dataset can have a significant impact on the accuracy of the output.

Author [8] also used a ML model to predict the AQI, however their results for the random forest regressor were only 80% accurate, but in our machine learning model have given the 99.87% accuracy. because we used the three primary feature engineering techniques and our model performed well on the dataset. The quality of that training dataset has increased.

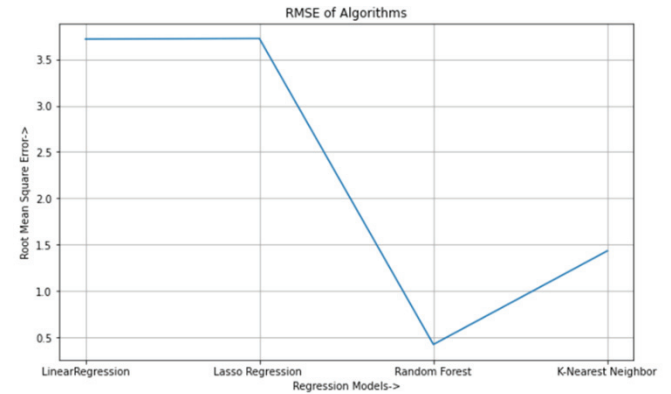


Fig. 11: RMSE of Machine Learning Models

TABLE I. ML MODELS COMPARISON

Machine Learning Model	Accuracy (%)	MAE	MSE	RMSE
Linear Regression	89.23	2.983	13.839	3.720
Lasso Regression	88.95	2.968	13.882	3.725
K-Nearest Neighbor Regressor	92.65	0.359	2.053	1.43
Random Forest Regressor	99.87	0.09	0.178	0.422

V. CONCLUSION

The objective of this research is to construct regression models utilizing supervised machine learning algorithms to forecast air quality index (AQI) by analyzing four different air pollutants. To achieve the best results, three primary feature engineering techniques were implemented. The findings revealed that both the RF Regressor model and KNN regressor produced satisfactory results, but the random forest regressor provided the highest accuracy and lowest root-mean-square error (RMSE) compared to the KNN regressor. Therefore, this model can be used to predict the AQI and identify the AQI range.

Future research will focus on enhancing this proposed model by developing a deep learning-based prediction model based on the day.

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